

Technical Documentation: Eye Movement & Face Attention Tracker

Framework: MediaPipe Tasks (Python) & OpenCV

System Compatibility: Hardware-agnostic (Dell, Asus, Mac, and Generic USB Webcams)

1. Project Overview

The objective of this project was to develop a high-stability, real-time attention monitoring system. Unlike traditional tracking that relies on fixed pixel coordinates, this system uses **normalized landmark ratios** and **temporal smoothing** to ensure accurate performance across different screen resolutions and lighting conditions.

2. Core Features

A. Dynamic Gaze Tracking (Pupil Movement)

The system tracks the iris center (Landmarks 468, 473) relative to the eye boundaries.

- **Horizontal Logic:** Calculates the ratio of the pupil between the inner and outer corners of the eye.
- **Vertical Logic:** Monitors pupil height relative to the upper and lower eyelids.
- **Output:** Detects LEFT, RIGHT, UP, DOWN, and CENTER.

B. Stable Head Pose Estimation

To detect if a user is looking away from the screen, the system monitors face orientation.

- **Horizontal (Yaw):** Compares the nose tip position against the lateral facial boundaries (ears).
- **Vertical (Pitch):** Compares the nose tip against the forehead and chin boundaries.
- **Output:** Detects FACE: FRONT, LEFT, RIGHT, UP, and DOWN.

3. Technical Implementation Details

Anti-Flicker Mechanism (Smoothing)

One of the primary challenges in Computer Vision is "jitter" (rapidly flickering text). We solved this using **Temporal Buffering**:

- Every movement (Gaze and Head) is passed through a `smooth_value` function.

- This results in a "locked-in" UI feel where the status only changes when a deliberate movement occurs.

Movement	Type	Threshold
Gaze (H)	Horizontal	Left < 0.42
Face (H)	Yaw	Left < 0.45
Face (V)	Pitch	Up < 0.40

4. Setup & Requirements

Dependencies

Bash

```
pip install opencv-python mediapipe numpy
```

File Structure

1. `main.py`: The core logic and UI rendering engine.
2. `face_landmarker.task`: The trained AI model (provided by Google MediaPipe) used for 478-point landmark detection.

5. Visual Interface (UI)

The system provides a real-time HUD (Heads-Up Display) including:

- **Status Box**: A semi-transparent overlay showing eye and face states.
- **Visual Indicators**: Red dots rendered directly on the pupils to confirm tracking lock.
- **Focus Feedback**: Text color changes to **Green** when the user is "Focused" (Face Front + Gaze Center) and **Orange/Red** when distracted.